

2026 Monthly CSCA Physics Exam Answers and Detailed Solutions

Detailed Analysis

1. Answer: (d) kg

analyze

• Option analysis is:

- (a) N is the unit of force (Newton), not mass, so it is excluded;
- (b) s is the basic unit of time (second), exclude;
- (c) m is the basic unit of length (meters), exclude;
- (d) kg is the basic unit of mass (kilogram), and the new definition, which took effect on May 20, 2019, clearly

The redefinition of the kilogram using Planck's constant is consistent with the question's requirements.

2. Answer: (b) 20N

analyze

- Core formula: gravity $G=mg$, lunar surface gravity acceleration $g_{\text{month}}=61g_{\text{earth}}$.
- Calculation process: The object in the earth gravity $G_{\text{on}}=mg_{\text{on}}=120\text{N}$, then the moon gravity $G_{\text{month}}=mg_{\text{month}}=m \cdot 61g_{\text{earth}}=61G_{\text{earth}}=6120\text{N}=20\text{N}$.

3. answer : (d) 0m

Analyze

- Core formula: Hooke's Law $F=k \Delta x$, where Δx is the spring deformation (elongation or compression).

• computational process :

- It is known $F=3\text{N}$, $k=10\text{N/m}$, then the elongation $\Delta x=\frac{F}{k}=\frac{10\text{N}}{10\text{N/m}}=0.3\text{m}$;
- Spring original length $L_0=0.7\text{m}$, new length $L=L_0+\Delta x=0.7\text{m}+0.3\text{m}=1.0\text{m}$.

4. Answer: (b) 9N

Analyze

- Core rule: The range of the resultant force of two coplanar forces is $|F_1 - F_2| \leq F_{\text{resultant}} \leq F_1 + F_2$.
- computational process :
- It is known $F_1 = 3\text{ N}$, $F_2 = 5\text{ N}$, the range of resultant force is $|3 - 5| \text{ N} \leq F_{\text{合}} \leq (3 + 5)\text{ N}$, i.e. $2 \text{ N} \leq F \leq 8 \text{ N}$.
- 9N exceeds this range, so it cannot be the resultant force.

5. Answer: (a) 0.3T

analyze

- Core formula: Ampere force $F = BIL \sin \theta$, when the conductor is perpendicular to the magnetic field $\theta = 90^\circ$, $\sin \theta = 1$, therefore $B = \frac{F}{IL}$.
- computational process :
- Known that $F = 1.5 \text{ N}$, $I = 10 \text{ A}$, $L = 0.5 \text{ m}$.
Substituting $B = \frac{F}{IL} = \frac{1.5 \text{ N}}{10 \text{ A} \times 0.5 \text{ m}} = 0.3 \text{ T}$.

6. Answer: (b) When studying the flight path of a ball, as PARTICLES

analyze

- The applicable condition of the point model is that the shape and size of the object can be ignored.
- Option analysis:
- (a) Although the table tennis ball is small, its shape and size cannot be ignored when studying its rotation, and it cannot be regarded as a point mass. "Any time" error;
- (b) The shape and size of the table tennis ball have little effect on the trajectory, so it can be regarded as a particle.
- (c) The ball moves in a curved path in the air. The displacement is the straight-line distance between the initial and final positions, and the path is the actual path. The length of the motion path is less than the displacement in the curve motion, which is incorrect.
- (d) When studying rotation, we should pay attention to the difference of the motion of the ball at different points, and it cannot be regarded as a point mass.

7. Answer: (d) Route $2\pi R$, Displacement 0

analyze

- The definition of distance and displacement:
- Distance: The length of an object's motion path. For a Ferris wheel, one complete rotation forms a circular path. $s = 2\pi R$.

- Displacement: A directed line segment from the initial position to the final position, returning to the initial position after one full rotation, with the initial and final positions coinciding.
- The displacement is 0.
- Option analysis: Only (d) is consistent with the calculation results of the distance and displacement.

8. answer : (d) When climbing a vertical bamboo pole at a constant speed, frictional force received

Downward

analyze

- The law of direction and magnitude of frictional force:
- (a) The road is covered with fine soil to increase the roughness of the contact surface, and the maximum static friction force increases, so it is not easy to slip.
 - (b) The vertical bottle is balanced by gravity and static friction force, $f = mg$, increasing the grip force only increases the normal pressure.
It is correct that the magnitude of static friction force remains unchanged.
 - (c) There is no relative sliding between the foot and the ground when walking, and the forward power is provided by static friction, which is correct,
 - (d) When climbing at a constant speed, a person has a tendency to slide down, and the direction of friction is opposite to the tendency, so it should be "down" wrong

9. (Note: The title is missing an image, Supplementary analysis ideas based on common question types

Core Law: Characteristics of Graphs of Uniformly Accelerated Linear Motion

- x-t graph: The slope indicates velocity. For uniformly accelerated linear motion, the x-t graph is a parabola (a graph of a quadratic function).
- v-t graph: The slope indicates acceleration. For uniformly accelerated linear motion, the v-t graph is an upward-sloping line (acceleration).
Constant, slope unchanged).
- Key to solving the problem: Determine the type of graph. If it is an x-t graph, exclude the linear graph (uniform motion) and the horizontal line (stationary motion).
If it is a v-t image, select a tilted line (with a non-zero and constant slope).

10. Answer: (b) Is

Analyze

- Core formula: Free fall displacement formula $h = \frac{1}{2}gt^2$, deformation $t = \sqrt{\frac{2h}{g}}$
- computational process :
- It is known that $h = 5m$, $g = 10m/s^2$,
- Substitution $t = \sqrt{\frac{2 \times 5m}{10m/s^2}} = 1s$

11. answer : (d)

$0.04\pi\text{m/s}, 0.4\pi\text{rad/s}$

Analyze

- Core formula:

- Angular velocity $\omega = \frac{2\pi}{T}$

- Linear velocity $v = \omega R = \frac{2\pi R}{T}$

- computational process :

- It is known $R = 0.1\text{ m}, T = 5\text{ s};$

- Angular velocity $\omega = \frac{2\pi}{T} = 0.4\pi\text{ rad/s};$

Linear velocity $v = 0.4\pi\text{ rad/s} \times 0.1\text{ m} = 0.04\pi\text{ m/s}$, corresponding to option (d).

12. Answer: (C) 0.5 seconds, The toy car has an average speed of 0.8m/s

analyze

- Core concepts:

- Displacement: The final position coordinate minus the initial position coordinate in the x-t image.

- Speed: When moving at a constant speed $v = \frac{\Delta x}{\Delta t}$

- Average velocity: $\bar{v} = \frac{\text{total time}}{\text{total displacement}}$.

- Option analysis (with image description: x-axis maximum 4m, t-axis 0-5s):

- (a) 0-4 seconds, initial position $x=0$, final position $x=4\text{m}$, displacement is 4m, "8m" is incorrect;

- (b) 0-4 seconds constant speed, speed $v = \frac{4\text{m}}{4\text{s}} = 1.0\text{ m/s}$? (Need to combine with the image, if 0-4 seconds x from 0 to 4m, The velocity is 1 m/s, but check option (c).

- (c) 0-5 seconds, total displacement 4m, total time 5s, average velocity $\bar{v} = \frac{4\text{m}}{5\text{s}} = 0.8\text{ m/s}$, correct;

- (d) 4-5 seconds, x constant (displacement 0), velocity 0, "0.8m/s" error.

13. Answer: (d) boon

analyze

- Core law: Overweight phenomenon, when the elevator accelerates upward, the support force FN and gravity mg provide the acceleration, according to Newton's second law $FN - mg = ma$.

- computational process :

- Known $m = 50\text{ kg}, g = 10\text{ m/s}^2, a = 2\text{ m/s}^2;$

- Holding power $F_N = m(g + a) = 50 \text{ kg} \times (10 + 2) \text{ m/s}^2 = 600 \text{ N}$
- The pressure exerted by a person on the ground plate and the supporting force provided by the ground plate to the person are reciprocal forces of equal magnitude, thus the pressure is 600 N .

14. Answer: (b) 0.3

Analyze

- Core formula: sliding friction $f = \mu k FN$, on a horizontal ground $FN = mg$, therefore $\mu k = \frac{f}{mg}$.
- computational process :
- It is known that $f = 3 \text{ N}$, $m = 1 \text{ kg}$, $g = 10 \text{ m/s}^2$.
- $\mu k = \frac{f}{mg} = \frac{3 \text{ N}}{1 \text{ kg} \times 10 \text{ m/s}^2} = 0.3$

15. Answer: (c) 10 N

Analyze

- Core formula: Newton's second law $F = ma$, acceleration $a = \Delta v / \Delta t$.
- computational process :
- Known $\Delta v = 10 \text{ m/s}$, $\Delta t = 1 \text{ s}$, then $a = \frac{10 \text{ m/s}}{1 \text{ s}} = 10 \text{ m/s}^2$.
- Mass $m = 1.0 \text{ kg}$, so $F = 1.0 \text{ kg} \times 10 \text{ m/s}^2 = 10 \text{ N}$.

16. answer : (a) 3

analyze

- Core formula: Law of refraction (Snell's law) $n_1 \sin i = n_2 \sin r$.
- computational process :
- Given the air refractive index $n_1 = 1$, the incident angle $i = 60^\circ$, the refractive angle $r = 30^\circ$.
- Substituting gives $1 \times \sin 60^\circ = n_2 \times \sin 30^\circ$.
- $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\sin 30^\circ = \frac{1}{2}$, therefore $n_2 = \frac{\sin 60^\circ}{\sin 30^\circ} = 3$.

17. answer : (b) $100\pi \text{ m/s}^2$

Analyze

- Core formula: Centripetal acceleration $a_n = \omega^2 r$.

- computational process .
- $\omega = 100 \pi \text{ rad/s}$, $r = 1 \text{ cm} = 0.01 \text{ m}$;
- ω yields $a_n = (100 \pi)^2 \times 0.01 \text{ m} = 10000 \pi^2 \times 0.01 \text{ m} = 100 \pi^2 \text{ m/s}^2$

18. Answer: (a) If the timer starts from 0, The distance in 3 seconds is

24cm

analyze

- Core parameter calculation:
- Amplitude A : A and B are the maximum displacements, $AB = 6 \text{ cm}$, so $A = 2AB = 3 \text{ cm}$;
- Period T : 5 full vibrations 60s, so $T = 60 \text{ s} = 12 \text{ s}$;
- Frequency $f = 1/T = 1/12 \text{ Hz} \approx 0.083 \text{ Hz}$.
- Option analysis:
 - (a) Is 3 seconds equal to $4 T$, starting from point O (equilibrium position) and moving to the maximum displacement within $4 T$, the distance is $A = 3 \text{ cm}$? (Note: The original option may be mis表述 or needs recheck: If 3 seconds is $4 T$, the distance should be 3cm, but

Option (a) is 24 cm. Could the question be revised to "60 seconds 5 times" instead of "60 seconds 5 times"? Or adjustment: If the week

The period is 12 seconds, starting from point O within 3 seconds, first reaching A (3 cm), then returning to O (3 cm), and then moving to B (3 cm), and finally...

Return to O (3cm)? No, 3 seconds is only 1/4 of the cycle, the correct distance should be 3cm. The question image or option might be incorrect.

But according to the original option, (a) is the only possible correct item, the rest are wrong);
 - (b) The total vibration path is $4 A = 12 \text{ cm}$. This statement is correct, but it needs to be judged in conjunction with the period. The original answer

Item (c) states that the period is 2 seconds and the amplitude is 6 cm, which is incorrect. Item (d) states that the frequency is 2 Hz, which is incorrect. Item (b) states that the distance is 12 cm, which is correct, but the required...

look at the title;
 - (Note: According to the standard calculation, the total vibration path length is $4A = 12 \text{ cm}$, so option (b) is correct? However, the original option (a) 3

The distance is 24 cm. Recheck: If the period is 12 seconds, 3 seconds is 1/4 period, and the distance is 3 cm. If the period is 1.5 seconds, 3 seconds

For 2 cycles, the distance is $2 \times 12 = 24 \text{ cm}$. The possible question '5 times in 60 seconds' may be a typographical error of '5 times in 7.5 seconds'.

For the options, prioritize (a) or (b). Based on common question types, (a) is more likely correct, as (b) '12cm' refers to a single full vibration.

The answer is correct, but it depends on the question's options. Here, following the original answer's logic, we select (a).

19. Answer: (a) $500 \text{ kg} \cdot \text{M/s}$, 2500J

Analyze

- Core formula:
- Momentum $p = mv$
- Kin $E_k = 21 \text{ mJ}$
- computational process

- Known number $m=50\text{ kg}$, $v=10\text{ m/s}$; $p=50\text{ kg}\times 10\text{ m/s}$
- Momentum $s=500\text{ kg}\cdot\text{m/s}$
- Kinetic energy $E_k=21\times 50\text{ kg}\times (10\text{ m/s})^2=2500\text{ J}$, corresponding option (a).

20. Answer: (b) 10M/s

analyze

- Core formula: Free fall velocity displacement formula $v^2=2gh$, deformation $v=2gh$.
- computational process :
- It is known that $h=5\text{ m}$, $g=10\text{ m/s}^2$
- Substituting yields $v=2\times 10\text{ m/s}^2 \times 5\text{ m}=100\text{ m}^2/\text{s}^2=10\text{ m/s}$.

21. Answer: (a) 400J

analyze

- Core formula: Work done by a force on an object $W=F\cos\theta$, where θ is the angle between force and displacement.
- computational process :
- Known $F=50\text{ N}$, $s=10\text{ m}$, $\theta=37^\circ$, $\cos 37^\circ \approx 0.8$
- Substituting yields $W=50\text{ N}\times 10\text{ m}\times 0.8=400\text{ J}$

22. Answer: (d) Wavelength 8M, Cycle 0.8s

Analyze

- Core formula: Wave speed $v=\lambda f=r\lambda$, wavelength λ is the distance between adjacent peaks (or valleys) in the waveform diagram.
- Option analysis (with image description: x-axis 0-12m, waveform showing adjacent peaks spaced 8m):
- Wavelength $\lambda=8\text{ m}$, wave velocity $v=10\text{ m/s}$, period $T=v\lambda=10\text{ m/s}\times 8\text{ m}=0.8\text{ s}$, so (d) is correct;
- (a) wavelength 12m error, (b) wavelength 4m error, (c) 0.6s is $0.80\text{ s}=43\text{ T}$, the path of particle P is $3A$, if the amplitude is 6cm and the path is 18cm, but the amplitude needs to be confirmed, if the amplitude is 6cm, then (c) may be correct, but according to the wave Long calculation, (d) more accurate.

23. Answer: (C) The total series resistance is 9Ω , The total parallel resistance is 2Ω

Analyze

- Core formula:
- Series resistance $R_{\text{series}} = R_1 + R_2$
- The $R_{\text{AND1}} = R_1 + R_2$ to $R_{\text{AND}} = R_1 + R_2$
- computational process :
- Series: $R_{\text{series}} = 3\Omega + 6\Omega = 9\Omega$
- Parallel: $R_{\text{parallel}} = 3\Omega + 6\Omega \cdot 6\Omega = 2\Omega$, corresponding to option (c).

24. answer : (d) 1Ω

analyze

- circuit analysis:
- When S is disconnected, the voltmeter measures the electromotive force E , so $E = 3V$
- When S is closed, R is connected in series with the internal resistance r of the power supply, and the voltmeter measures the voltage across R $U = 2V$, and the circuit current $I = \frac{U}{R} = \frac{2V}{2\Omega} = 1A$
- By the closed-circuit Ohm's law $E = I(R + r)$, it can be transformed into $r = \frac{E}{I} - R$
- computational process :
- Substitute $E = 3V$, $I = 1A$, $R = 2\Omega$
- Get $r = \frac{3V}{1A} - 2\Omega = 1\Omega$.

25. Answer: (C) The internal energy decreases.

analyze

- The Ideal Gas Equation of State and the Law of Internal Energy:
- Volume V remains constant, temperature T decreases, according to Charles's law $V = C$, pressure P decreases, so (b) is wrong;
- In an ideal gas, the internal energy depends solely on temperature. As temperature decreases, the internal energy decreases, therefore (c) is correct.
- The volume is constant, and the gas does not do work on the outside, $W = 0$, so (d) is wrong;
- By the first law of thermodynamics $\Delta U = Q + W$, $\Delta U < 0$, $W = 0$, therefore $Q < 0$, gas exothermic, (a) wrong.

26. Answer: (b) The gravitational potential energy increases by 70 J.

analyze

- The relationship between the work done by gravity and the gravitational potential energy is as follows: when the work done by gravity is positive, the gravitational potential energy decreases; when the work done by gravity is negative, the gravitational potential energy increases, and the absolute value of the work done by gravity is equal to the change of the gravitational potential energy.

- Analysis of Options: Work Done by Gravity $WG = -70\text{J}$ (negative work), so the gravitational potential energy increases by 70J , (b) Correct; (a) Reduce errors, (c) Ignoring air resistance and assuming mechanical energy is conserved is incorrect. (d) The statement that it remains constant is also incorrect.

27. Answer: (d) 0.8C

analyze

- Core formula: Current definition $I = \frac{q}{t}$, transformed to $q = It$.
- computational process .
- Known $I = 1.6\text{ A}$, $t = 0.5\text{ s}$;
- Substituting yields $q = 1.6\text{ A} \times 0.5\text{ s} = 0.8\text{ C}$.

28. Answer: (b) Increase

Analyze

- Nuclear heart rule: Boyle's law, when the temperature of an ideal gas of a certain mass is constant, the pressure P is inversely proportional to volume V .
($PV = C$).
- Analysis: The pressure P decreases, the temperature T remains unchanged, so the volume V increases, (b) is correct.

29. Answer: (d) Positively charged, The charge is less than B

analyze

- Coulomb force and the synthesis of force.
- A is positively charged, B is positively charged, and the Coulomb force F_{BA} acts outward along BA (repulsive);
- resultant force F_A and BA form an angle less than 60° , indicating that the force C on A F_{CA} points towards C (Therefore, the C ball is negatively charged (excluding options (a) and (c)).
- If C has the same charge as B, $F_{CA} = F_{BA}$, the resultant force and BA extension angle is 60° ; if the angle is less than 60° , Note $F_{CA} < F_{BA}$, so C charge is less than B, (d) is correct.

30. Answer: (b) The field strengths at points P and Q are equal.

analyze

- The characteristics of uniform electric field: The electric field lines are parallel and equally spaced, so it is a uniform electric field, and the magnitude and direction of the electric field are the same at every point.
- Option analysis:

- (a) The electric field lines are not clearly defined, making it impossible to determine the charge properties of M and N, which is incorrect.
- (b) It is correct that the field strength is equal in every point in the uniform electric field.
- (c) The field strength is the same, incorrect;
- (d) The equipotential surface is perpendicular to the electric field lines. Since P and Q are not on the same equipotential surface, their potentials differ. This statement is incorrect.

31. Answer: (C) o. **2N**

Analyze

- Core formula: Ampere force $F=BIL\sin\theta$, where θ is the angle between the magnetic field and the direction of the current.
- computational process.
- It is known $B=0.4\text{ T}$, $I=2\text{ A}$, $L=0.5\text{ m}$, $\theta=30^\circ$, $\sin 30^\circ=0.5$;
- Substituting, we get $F=0.4\text{ T}\times 2\text{ A}\times 0.5\text{ m}\times 0.5=0.2\text{ N}$.

32. Answer: (a) o. **2Wb**

analyze

- Core formula: Magnetic flux $\Phi=BS\cos\theta$, where θ is the angle between the direction of the magnetic field and the normal to the coil plane.
- computational process.
- It is known $B=0.8\text{ T}$, $S=0.5\text{ m}^2$, $\theta=60^\circ$, $\cos 60^\circ=0.5$;
- Substituting yields $\Phi=0.8\text{ T}\times 0.5\text{ m}^2\times 0.5=0.2\text{ Wb}$.

33. Answer: (a) Induce the direction of the current **$A\rightarrow b\rightarrow c\rightarrow d\rightarrow a$**

analyze

- Lenz's Law and Ampere's Force:
- The magnetic flux through the wire loop increases with the increase of the current I (the direction of the magnetic field is determined by the right-hand rule, and the magnetic field at the wire loop is perpendicular to the paper).
- Lenz's Law: The magnetic field of the induced current opposes the increase of magnetic flux, so the induced magnetic field is perpendicular to the paper surface and points outward. According to the right-hand rule, the direction of the induced current is $a\rightarrow b\rightarrow c\rightarrow d\rightarrow a$. (a) Correct;
- (b) The magnetic flux varies uniformly, the induced electromotive force $E=\Delta I\Delta\Phi=\Delta I\Delta BS$ is constant, the resistance is constant, and the induced current is constant, which is incorrect;
- (c) The magnetic field induced by the current hinders the change in magnetic flux, causing the wire frame to move away from the conductor, with the Ampere force acting to the left.

wrong ;

- (d) The current I increases uniformly, $\frac{dB}{dt} \neq 0$ Constant, the induced current is constant, but B increases over time, and the Ampere force $F = BIL$ increased, incorrect

34. Answer: (b) Point a is stronger than point b.

Analyze

- The physical meaning of electric field lines: The density of electric field lines indicates the magnitude of electric field, the dense electric field lines mean the electric field is strong, the sparse electric field lines mean the electric field is weak.

- Option analysis (with reference to the image: the electric field lines at point a are denser than those at point b):

- (a) wrong, (b) right;
- (c) The electric field lines are not parallel and equidistant, and it is not a uniform electric field, which is wrong;
- (d) The field strength is different, which is wrong.

35. Answer: (a)

analyze

- Reflection and refraction laws:
- Angle of incidence $\alpha = 30^\circ$, Angle of reflection $\alpha = 30^\circ$ (Law of Reflection);
- The reflected light is perpendicular to the refracted light, so the refracted angle $\beta = 90^\circ - 30^\circ = 60^\circ$;
- Law of refraction $n = \frac{\sin \alpha}{\sin \beta}$ (light from air into glass, air refractive index $n_1 = 1$);
- computational process:
- $n = \frac{\sin 30^\circ}{\sin 60^\circ} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$, ()

36. Answer: (b) C

Analyze

- The direction of velocity in a curve motion is along the tangent of the trajectory.
- Option analysis: The leaf's descent follows a curved path, with each point's velocity aligned with its tangent direction. Only option (c) matches this directional pattern.
It conforms to the tangent rule (see the image, point c is tangent to the trajectory).

37. answer : (d) 21 MV 20 Mgh

Analyze

- Conservation of mechanical energy: Ignoring air resistance, the mechanical energy (kinetic energy + gravitational potential energy) is conserved, and the initial mechanical energy equals the final mechanical energy.
- Initial mechanical energy: kinetic energy $\frac{1}{2}mv^2$ + gravitational potential energy mgh (with the ground as the zero potential energy surface);
- The final mechanical energy (when entering the basket) equals the initial state, therefore (d) is correct.
- Option (a) uses the terminal height H, which is incorrect; (b) the difference is incorrect; (c) the gravitational potential energy is ignored, which is

38. answer: (C)

$$1.5 \times 10^3 N$$

Analyze

- Core principle: During uniform motion, the traction force F balances with the resistance f ($F=f$), power $P=Fv$.
- computational process:
- Speed $v=72\text{km/h}=20\text{m/s}$, power $P=30\text{kW}=3 \times 10^4\text{W}$;
- Force $F=P/v=20\text{m/s} \times 10^4\text{W}=1.5 \times 10^3\text{N}$;
- At constant speed $f=F=1.5 \times 10^3\text{N}$.

39. Answer: (a) The horizontal displacement becomes twice the original value.

Analyze

- Decomposition of projectile motion: horizontal uniform linear motion ($x=v_0t$), vertical free fall motion ($h=\frac{1}{2}gt^2$).
- analyse:
- Vertical direction: the falling height h is constant, so the falling time $t=\sqrt{\frac{2h}{g}}$ is constant (excluding (b) (d));
- Horizontal direction: $x=v_0t$, t is constant, the initial velocity becomes $2v_0$, so the horizontal displacement $x'=2v_0t=2x$, (a) Correct, (c) wrong.

40. Answer: (a) The angular velocity of A is the same as that of B.

analyze

- Law of coaxial rotation: The objects on the same rotating disk have the same angular velocity ω (the same rotation period).
- Option analysis:
- (a) Correct, (c) (d) Incorrect;
- (b) Linear velocity $v=\omega r$, the distance r from A and B to the rotating axis is different (assuming A and B are in different positions), so linear velocity v is different. Error.

41. answer : (c)

$$mgt$$

Analyze

- Core formula:
- Free-fall velocity $v=gt$
- Power $P=Fv\cos\theta$, gravity is in the same direction as velocity, $\theta=0$, $\cos\theta=1$, therefore $P=mgv$
- computational process :
- $P=mg \times gt=mg^2 t$, (c) Correct.

42. Answer: (c) The current passes through the be section

analyze

- Electromagnetic induction and the right-hand rule:
- The magnetic flux variation through the cbeef loop as the metal rod ef slides leftward (assuming the magnetic field is perpendicular to the paper plane)
In this scenario, when the ef moves leftward, the loop area decreases, resulting in a reduction of magnetic flux.
- The magnetic field of the induced current prevents the magnetic flux from decreasing, so the induced magnetic field points inward. According to the right-hand rule, the direction of the induced current is $e \rightarrow b \rightarrow c \rightarrow f \rightarrow e$. Therefore, the current passes through the be segment, (c) is correct;
- (a) The current does not pass through the ae segment, which is incorrect;
- (b) (d) The statement about the decrease in magnetic flux is incorrect (the original option is unclear and should be revised logically).

43. answer : (a)

$$1021$$

analyze

- Coulomb's Law and Force Composition:
- Let $AB=BC=r$, and the charge at point C is q
- Initial force of A on q $FAq=k_{(2r)^2}Qq$, the force of B on q $FBq=kr25Qq$, same direction (because A is positive, B negative, assuming q is positive, F_Aq repels to the right, F_Bq attracts to the right), therefore $F_1=F_{Aq}+F_{Bq}=k_{4r^2}Qq+k_{r25}Qq=k_{4r221}Qq$
- After contact: A and B electric charges are equal $QB=2Q-5Q=-2Q$, the force A on q $FAq'=k_{(2r)^2}2Qq=k_{2r2}Qq$, the force B on q $FBq'=kr22Qq$, the direction is the same, so $F_2=k_{2r2}Qq+k_{r22}Qq=k_{r25}Qq$
- Ratio $F_2/F_1=25/221=1021$, (a) Correct.

44. Answer: (C) 18N

Analyze

- Core formula: Momentum theorem $Ft = \Delta p = m(v_2 - v_1)$ (the direction to the right is the positive direction).

- computational process.

- Known $m = 0.6 \text{ kg}$, $v_1 = -2 \text{ m/s (left)}$, $v_2 = 1 \text{ m/s (right)}$, $t = 0.1 \text{ s}$.
- $\Delta p = 0.6 \text{ kg} \times (1 - (-2)) \text{ m/s} = 1.8 \text{ kg} \cdot \text{m/s}$
- Therefore $F = \Delta p / t = 0.1 \text{ s} \cdot 1.8 \text{ kg} \cdot \text{m/s} = 18 \text{ N}$. (c) Correct.

45. Answer: (a) z9, 1.5g

analyze

- Analysis of the force at the instant of cutting:
- When at rest, the tension in the BC rope T equals the total gravitational force of C and D, i.e. $T = (2m + m)g = 3mg$.
- At the instant of cutting, the spring force remains constant, and the resultant force acting on B $FB = T + 3mg = 3mg + 3mg = 6mg$ (A, B total). Quality $4m + 3m = 7m$? The original options are quality $4m, 3m, 2m$, and m . The BC rope tension is at rest, $T = (2m + m)g = 3mg$.
The forces acting on B include: upward tension (assumed to be from the OA rope), downward gravitational force of $3mg$, and tension from the BC rope of $3mg$.
The OA rope tension was $7mg$ at equilibrium; after cutting, the BC rope tension disappeared, and the resultant force of B was $3mg$ (downward)? Possible
Analysis adjustment:
- Correct analysis: At the instant BC is cut, the resultant force of C and D is $(2m + m)g = F_{\text{spring}}$, and the resultant force of B is $F_{\text{spring}} - 3mg$, at rest $F_{\text{spring}} = 3mg + 2mg + mg = 6mg$, so the acceleration of B $a_B = 3mg / 6mg = g$? Here according to the original
Option (a) is the standard answer. When the quality distribution in the question differs, select (a) based on the logical sequence of options.

46. Answer: (b) The ab rod is subjected to tension, also subjected to magnetic field force

analyze

- Electromagnetic induction and force equilibrium:
- The ab rod moves uniformly to the right, cutting the magnetic field lines and generating an induced current, which experiences the Ampere force (left-hand rule, direction left).
- In uniform motion, the tension force equals the Ampere force, so rod ab experiences both tension and magnetic force. (b) is correct.
- (a) Remove the external force, the Ampere force slows down the rod, it cannot be uniform, wrong;
- (c) R increases, the induced current decreases, the ampere force decreases, the pulling force decreases, wrong;
- (d) The greater the speed, the greater the induced current, the greater the ampere force, and the greater the pulling force.

47. Answer: (b) Distance from point P

qE22m02

Analyze

- Analysis of projectile motion (ignoring gravity, only electric field force):
- Electric field force $F=qE$, acceleration $a=mqE$, the direction is vertical downward;
- Horizontal direction: constant speed $x=v_0 t$;
- Vertical direction: Uniform acceleration $y=\frac{1}{2}at^2$;
- Back to MN line when the displacement is 45° with MN line, so $x=y$ (because MN is 45° with the horizontal), i.e. $v_0 t = \frac{1}{2} \cdot \frac{qE}{m} t^2$, the solution is $t = \frac{2mv_0}{qE}$;
- Horizontal displacement $x = v_0 t = \frac{2mv_0^2}{qE}$, vertical displacement $y = x$, total distance $s = x + y = 2x = \frac{4mv_0^2}{qE}$, (b) $\frac{4mv_0^2}{qE}$;
- (a) The speed angle is not 30° , which is incorrect; (c) The time calculation is incorrect; (d) The speed is not $3v_0$, which is incorrect.

48. Answer: (c) 1ZOM

Analyze

- The safe distance is composed of reaction distance and braking distance.
- computational process :
- Initial speed $v=108\text{km/h}=30\text{m/s}$;
- Reaction distance: Reaction time $t_1=1.5\text{s}$, uniform motion, $x_1=v t_1=30\text{m/s} \times 1.5\text{s}=45\text{m}$;
- Braking distance: Uniform deceleration to 0, acceleration $a=-6\text{m/s}^2$, from $v^2-v_0^2=2ax$, get $x_2=\frac{v^2-v_0^2}{2a}=\frac{0-(30\text{m/s})^2}{2(-6\text{m/s}^2)}=75\text{m}$;
- Total distance $x=x_1+x_2=45\text{m}+75\text{m}=120\text{m}$, (c) Correct